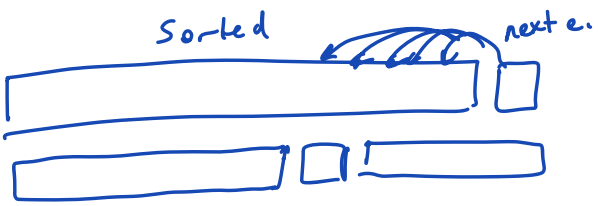
Insertion Sort:

Best case : array is already sorted.

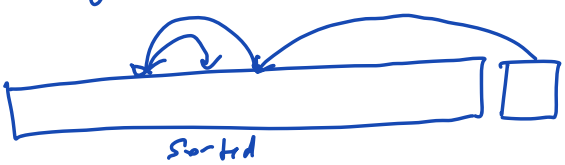
Worst case : array is reversely ordered.

$$C_{Best}(n) = \sum_{i=1}^{n-1} 1 = n-1 \in \Theta(n)$$

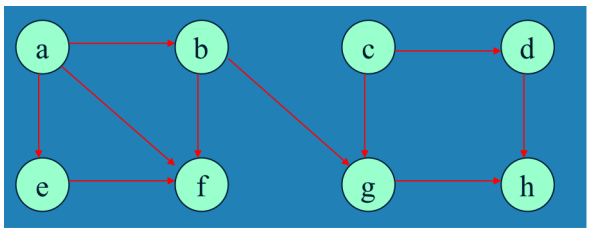
$$C_{worst}(n) = \sum_{i=1}^{n-1} \sum_{j=0}^{i-1} 1 = \sum_{i=1}^{n-1} i = \frac{(n-1)n}{2} \in \Theta(n^2)$$



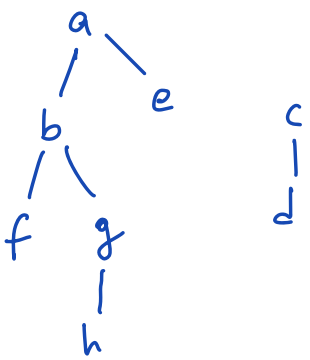
Binary insertion sort



We apply binary search to find the correct place to insert the next element.



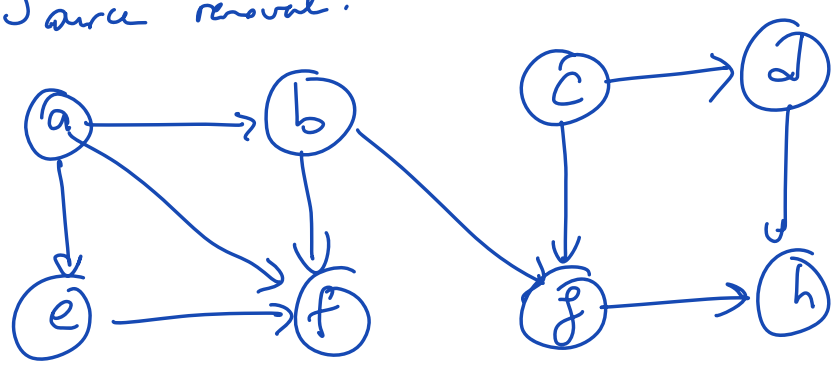
$h_{5,2}$
$f_{3,1}$ $g_{4,3}$
$b_{2,4}$ $e_{6,5}$ $d_{8,7}$
$a_{1,6}$ $c_{7,8}$



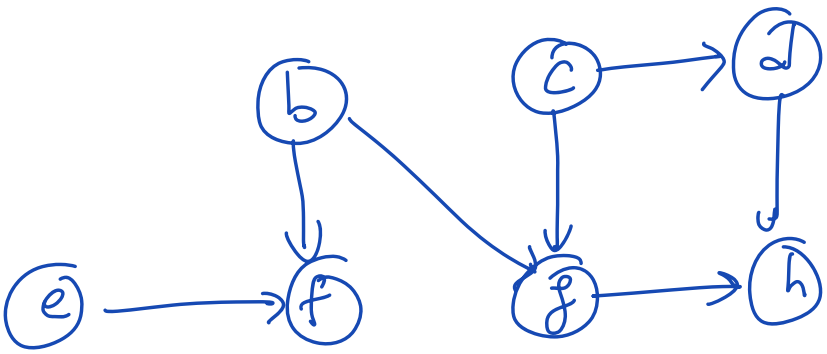
pop order: f, h, g, b, e, a, d, c

reverse pop order: c, d, a, e, b, g, h, f.
→ topological order.

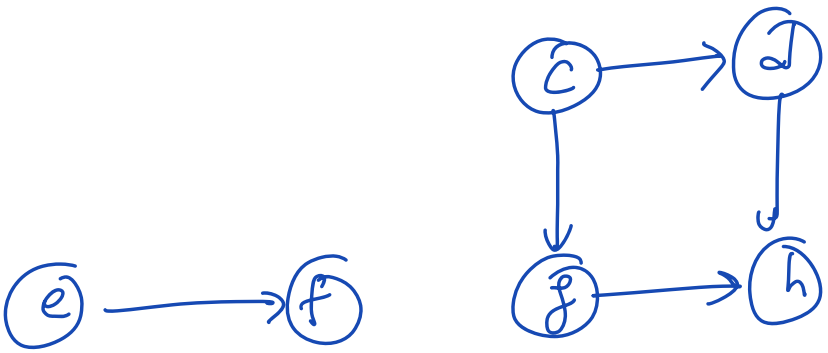
Source removal.



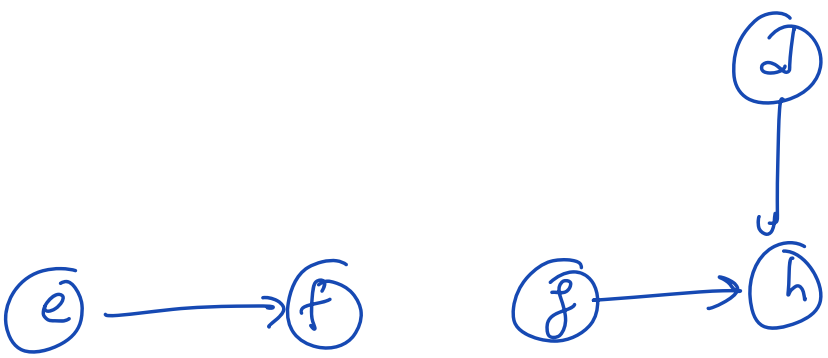
sources:
a, c
remove: a



sources:
e, b, c
remove: b.



src:
e, c
remove c



src:
d, g, e
remove d.



remove e
f
g
h

order of reveal: a b c d e f g h

→ topological sort.

Generating permutations

1 2
2 1

12 21

123 132 312
321 231 213

1234 1243 1423 4123
4132 1432 1342 1324
⋮

Generating Subsets:

{a, b, c, d, e}
0 0 1 0 1 ⇒ {c, e}
1 0 0 1 1 ⇒ {a, d, e}

n = 1 0 1
n = 2 0 0 0 1
1 1 1 0

L1: 0 1
L2: 1 0

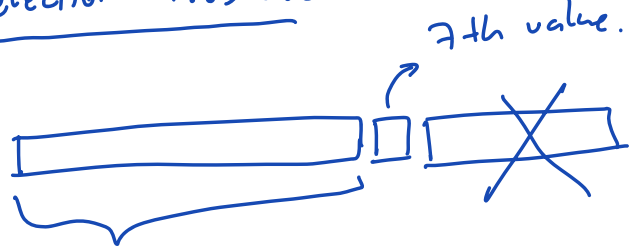
n=3 : L1: 000 001 011 010

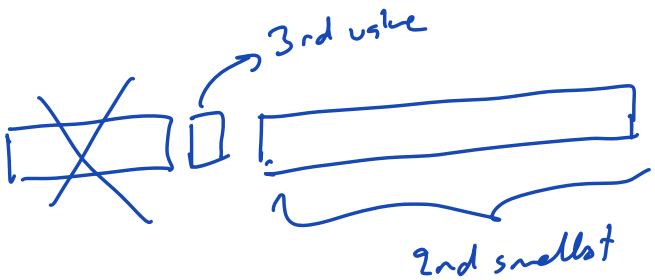
L2: 110 111 101 100

n=4 L1: 0000 0001 0011 0010 0110 0111 0101 0100

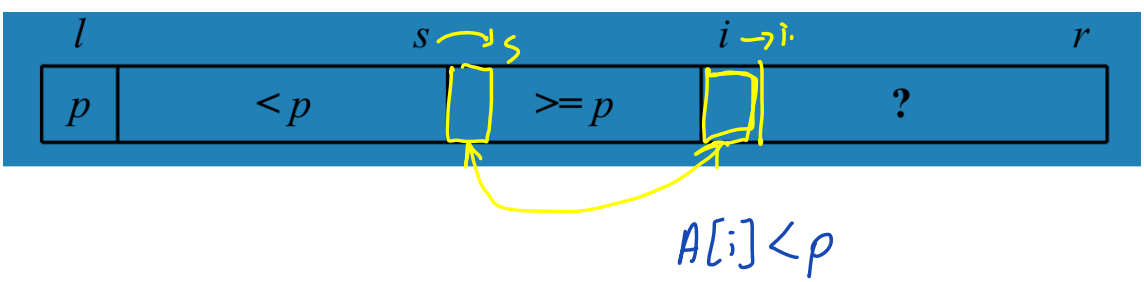
L2: 1100 1101 1111 1110 1010 1011 1001 1000

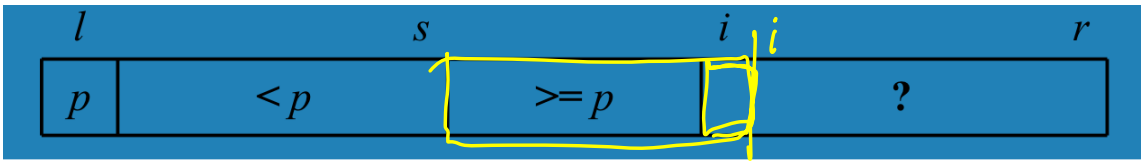
Selection Problem





Lomuto's partitioning.





$A[i] \geq p$

Worst case of interpolation search:

0 0.0001 0.0002 0.0003 ... - 0.001 10000